US Department of Agriculture (USDA)



Forest Service



Southern Research Station

Statement of Work

Laboratory (Site ID #02) HVAC Replacement
Santee Experimental Forest
3734 SC-402
Cordesville, SC 29434

Contents

General Information 3
1.0 Scope of Work 3
2.0 Background 3
Contractor Requirements 3
3.0 Technical Requirements/Tasks 3
3.1 HVAC Equipment Replacement 3
4.0 Materials 3
4.1 HVAC Equipment Replacement 3
5.0 Submittals 4
6.0 Government Furnished 4
7.0 Travel 4
8.0 Contractor's Key Personnel 4
9.0 Government's Key Personnel 4
10.0 Schedule 4
11.0 Security Requirements 5
12.0 Clean Up/Disposal 5
13.0 Data Rights 5
14.0 Section 508 – Electronic and Information Technology Standards 5
Attachment 8

Project
Santee Experimental Forest Laboratory (Site ID #02) HVAC Replacement

General Information

1.0 Scope of Work

- The Government requires the replacement of an existing 3-ton packaged heat pump system with a new, high-efficiency **ENERGY STAR-certified** unit that also satisfies **FEMP** federal purchasing requirements for energy-consuming products.
- The unit serves the Laboratory at the Santee Experimental Forest located at 3734 SC-402, Cordesville, SC 29434 (33.136543, -79.816041) https://maps.app.goo.gl/BwsJDgTcSUj3JfbD7

2.0 Background

A heat pump (HP3) serving a portion of the 2480 GSF LAB building at the Santee Experimental Forest has failed and requires replacement. See Figures 1 and 2 for more information.
No potentially hazardous materials are expected to be encountered during the execution of work under this Scope of Work.
Should the contractor encounter any additional suspected hazardous materials, stop work and contact COR immediately.

Contractor Requirements

3.0 Technical Requirements/Tasks

3.1 HVAC Equipment Replacement

Contractor shall furnish all labor, materials, equipment, tools, supervision, and incidentals to:

1. **Demolition & Recovery**
   - Lockout/tagout; recover refrigerant per EPA rules; disconnect and remove the existing packaged unit, thermostat, and related wiring/condensate; dispose of materials per environmental rules.
2. **Installation**
   - Provide and install one (1) **new 3-ton ENERGY STAR-certified single packaged heat pump** unit sized to existing ducts and electrical service; equipment must meet Section 3 efficiency requirements.
   - Electrical: new **non-fused disconnect** (as required), whips, OCPD verification, labeling; outdoor unit base/pad/anchorage per code and site conditions.
   - Controls: install **new programmable thermostat**; integrate low-voltage wiring; verify staging/defrost/back-up heat (if present).
   - All equipment shall be installed per AHRI/ASHRAE best practices to preserve rated **SEER2/HSPF2** performance
   - **Condensate management:** proper trap, slope, cleanout; safety float switch; route to approved location; weather-seal penetrations; restore finishes.
3. **Commissioning & Verification**
   - Start-up per manufacturer; verify airflow (CFM), external static, supply/return temps, and refrigerant performance; functional test heating/cooling, defrost and controls; document measured data.
4. **Training & Turnover**
   - Provide owner training on thermostat, operation, and routine maintenance; deliver O&M, warranty certificates, and as-built routing for line set and electrical one-line.

4.0 Materials

4.1 HVAC Equipment Replacement

Packaged Air Source Heat Pump: Energy Star or FEMP Designated Product,
Thermostat: 7 day programmable; Energy Star or FEMP-Designated Product
Equipment Pad: 3” minimum thickness, precast or cast-in-place concrete
Equipment Risers: Installer’s choice, must be able to support equipment weight and prevent slipping and minimize vibration during operation of the unit.

Federal Efficient Purchasing Requirements: Agencies must procure **ENERGY STAR-certified** or **FEMP-designated** products (42 U.S.C. § 8259b; FAR Part 23) unless written exceptions are justified; products may be presumed life-cycle cost-effective (10 CFR Part 436, Subpart C). [5] [6]

- **Efficiency Baselines**: Post-2023 **SEER2/HSPF2** national minimums apply; residential split heat pumps must meet **≥14.3 SEER2** and **≥7.5 HSPF2**. Regional installation compliance (including the Southeast) is required.

ENERGY STAR Energy Efficient Products | Products | ENERGY STAR
https://www.energystar.gov/products
Search for Energy-Efficient Products | Department of Energy:
https://www.energy.gov/eere/femp/search-energy-efficient-products

5.0 Submittals

Submit the requested information for the following material for approval prior to incorporation into the work. Allow 10 working days for review/approval.

- Air Source Heat Pump product data
- Thermostat product data

Submit the requested information for the following to be incorporated into the project file prior to submission of the final project invoice.

- EPA Section 608 technician credentials and refrigerant recovery logs.
- Commissioning Report (startup checklist, measured performance)
- As-Built documentation.
- Documentation of registration and enrollment necessary to activate manufacturer's standard warranty on all new equipment.

6.0 Government Furnished

Access to the work site during normal business hours (0730-1700 M-F, excluding Federal Holidays)
Parking and storage areas in the vicinity of the work area. Coordinate final locations with SPOC.
120V power supply in the vicinity of the work area. Contractor shall provide any required extension cords
Potable water supply in the vicinity of the work area. Contractor shall provide any required hoses.
Access to public restrooms

7.0 Travel

Travel costs associated with the work shall be included in the overall fixed price provided in response to this SOW.

8.0 Contractor's Key Personnel

Contractor's Key Personnel and associated key duties are noted below. Any proposed changes to Key Personnel shall be approved by the COR prior to implementation.
Site Superintendent – Contractor's onsite representative, directly involved in oversight of all work throughout the duration of the project and having the knowledge and authority to coordinate project-related decisions with the COR.
Installing Contractor shall be appropriately licensed; technicians **EPA 608-certified**; follow OSHA safety requirements and manufacturer instructions.

9.0 Government's Key Personnel

Contracting *Officer (CO): Chad Schmele chad.schmele@usda.gov*
Contracting *Officer's Representative (COR): Jason Anderson, Jason.L.anderson@usda.gov, 828-712-1923*
Site Point of Contact (SPOC): TBD

10.0 Schedule

The Contractor shall begin performance within 30 calendar days and complete it within 60 calendar days after receiving authorization to begin work from the Contracting Officer.

11.0 Security Requirements

Contractor's employees and subcontractors are required to check in with the site point of contact every day prior to beginning work. The SPOC will coordinate daily requirements for sign in and provide access to work areas as needed.

12.0 Clean Up/Disposal

Contractor shall provide daily removal of work-related debris and trash from the building to the disposal location.
Contractor shall provide final cleaning to restore workspace to original condition or better.
Onsite Dumpster is NOT available for use by the Contractor.

The Contractor is encouraged divert as much material as possible from disposal in a solid waste landfill by utilizing salvage businesses for the recyclable content generated from the work.

13.0 Data Rights

N/A

14.0 Section 508 – Electronic and Information Technology Standards

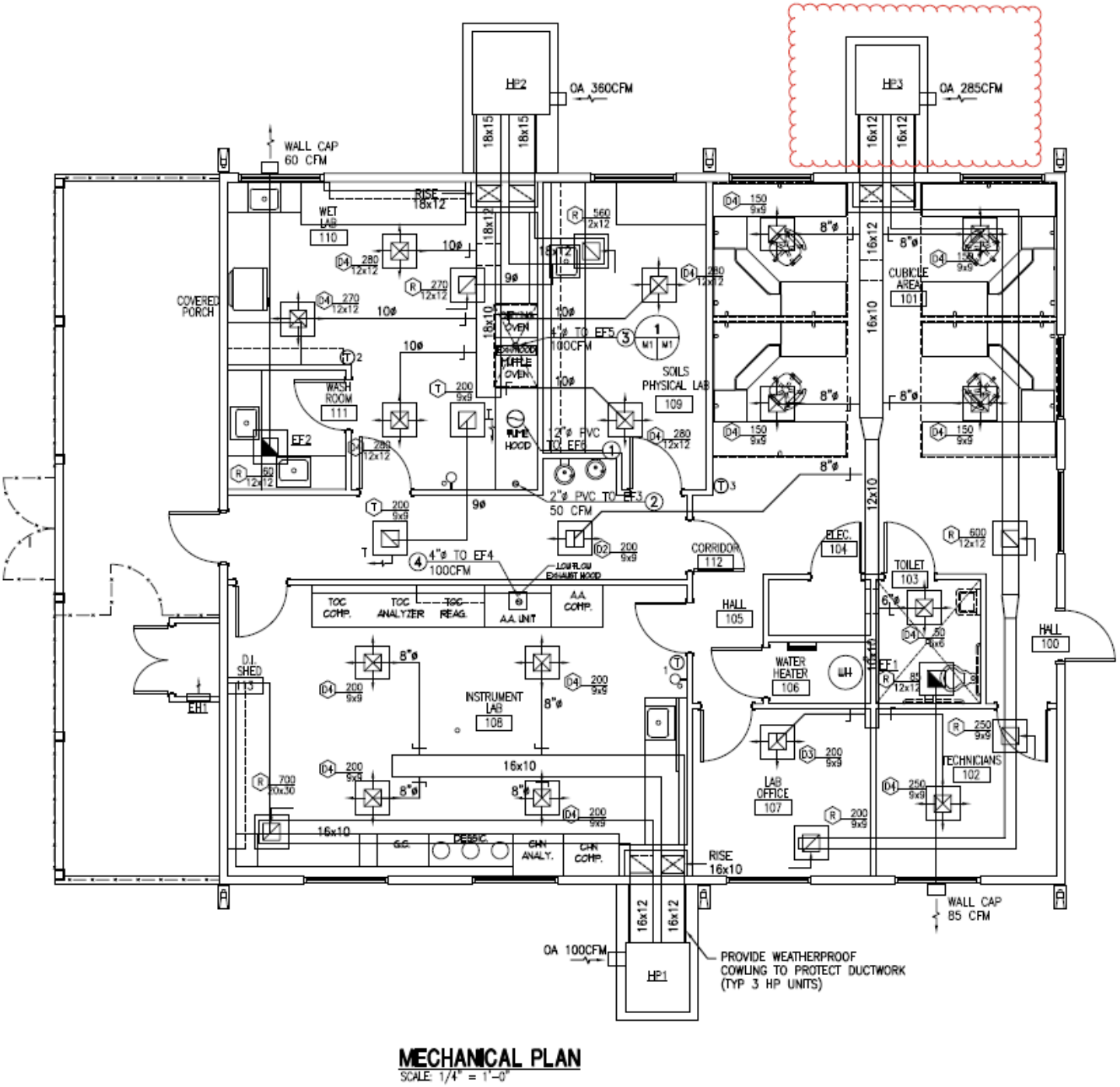
N/A

SINGLE PACKAGE HEAT PUMP SCHEDULE

			HP1	HP2	HP3
		SYMBOL			
		MANUFACTURER	CARRIER	CARRIER	CARRIER
		MODEL NO.	50SZ024301AD	50SZ042301AD	50SZ036301AD
		INDOOR (AHU)	–	–	–
		OUTDOOR (CU)	–	–	–
	SUPPLY AIR FLOW (CFM)		800	1400	1300
ELECTRICAL DATA	VOLTAGE	INDOOR	–	–	–
		OUTDOOR	208/230V, 1ø	208/230 V, 1ø	208/230 V, 1ø
	AUXILIARY HEAT		5 KW	10 KW	7.5 KW
	MIN. CIRC. AMPS	INDOOR	–	–	–
		OUTDOOR	40	56	56
	MAX. FUSE AMPS	INDOOR	–	–	–
		OUTDOOR	35	60	60
	OUTSIDE AIR CFM		100	360	285
	ACCESSORIES/NOTES		1,2,3, & 4	1,2,3, & 4	1,2,3, & 4

NOTES:
(1.) SINGLE PACKAGE HEAT PUMP
(2.) SINGLE POINT POWER CONNECTION.
(3) OUTDOOR AIR DAMPER
(4) FILTER RACK

Figure 1 - Schedule indicating design information for existing HP3



MECHANICAL PLAN
SCALE: 1/4" = 1'-0"

Figure 2 - Mechanical Plan



Figure 3 - Existing HP3 Thermostat



Figure 4 - Existing HP3 package unit



Figure 5 - Existing HP3 Nameplate Data



Figure 6 - HP3 package unit interior

Attachment